

BORIS ZUPANCIC

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<https://boriszupancic.github.io/>

EDUCATION	McGill University <i>Ph.D. in Mathematics</i> • Cumulative GPA: 4.00/4.00	Montréal, QC, Canada 2024 - 2028 (<i>expected</i>)
	Queen's University <i>M.Sc. in Mathematics</i> • Cumulative GPA: 4.12/4.30	Kingston, ON, Canada 2023 - 2024
	Queen's University <i>B.Sc. Honours in Mathematical Physics</i> • Cumulative GPA: 4.14/4.30	Kingston, ON, Canada 2019 - 2023
RESEARCH EXPERIENCE	Doctoral Research Thesis McGill University <i>In Poisson Geometry, Singularity Theory, Mathematical Physics.</i> • Topic/Title: <i>Resolution of Singularities in Poisson Geometry</i> • Advisor: Prof. Brent Pym	2024.09 - <i>Present</i>
	Master's Research Project Queen's University <i>In Riemannian and Hyperbolic Geometry, Dynamics.</i> • Topic/Title: <i>Marked Length Spectrum Rigidity of Surfaces</i> • Advisor: Prof. Thomas Barthelmé	2023.09 - 2024.08
	Bachelor's Research Thesis Queen's University <i>In Computational Astrophysics, Galactic Dynamics.</i> • Topic/Title: <i>Fuzzy Dark Matter Dynamics and the Quasi-Particle Hypothesis</i> • Advisor: Prof. Larry Widrow	2022.05 - 2023.08
TEACHING EXPERIENCE	MATH 314 - Advanced Calculus McGill University <i>Teaching Assistant: tutorials and grading.</i>	Fall 2025
	MATH 133 - Linear Algebra and Geometry McGill University <i>Teaching Assistant: tutorials and grading.</i>	Winter 2025
	MATH 150 - Calculus McGill University <i>Teaching Assistant: tutorials and grading.</i>	Fall 2024
	MATH 280 - Advanced Calculus Queen's University <i>Teaching Assistant: grading.</i>	Winter 2024
	MATH 225 - Ordinary Differential Equations Queen's University <i>Teaching Assistant: tutorials and grading.</i>	Fall 2023
	John Ursell Tutor Queen's University <i>Math Help Center Tutor: Calculus and Linear Algebra.</i>	Fall 2022 - Winter 2023
	PHYS 212 - Vibrations and Waves Queen's University <i>Teaching Assistant: grading.</i>	Winter 2022

AWARDS AND HONORS	• Doctoral Research Scholarship , Fonds de Recherches du Québec	2026 - 2028
	• Richard H. Tomlinson Doctoral Fellowship , McGill University	2024 - 2027
	• N. and G. Miller Graduate Fellowship , Queen's University	2023
	• John Ursell Tutor Award , Queen's University	2022
	• Dean's Honour List , Queen's University	2020 - 2023
	• Queen's University Excellence Scholarship , Queen's University	2019
PUBLICATIONS AND PREPRINTS	1. S. Lapointe, M. Matviichuk, B. Pym, and B. Zupancic. Weighted blowups and 3d Poisson desingularizations. Apr. 2026. doi: 10.48550/arXiv.2604.16698. URL http://arxiv.org/abs/2604.16698 . arXiv:2604.16698 [math]	
	2. B. Zupancic and L. M. Widrow. Fuzzy dark matter dynamics and the quasi-particle hypothesis. <i>Monthly Notices of the Royal Astronomical Society</i> , 527 (3):6189–6197, 11 2023. ISSN 0035-8711. doi: 10.1093/mnras/stad3620. URL https://doi.org/10.1093/mnras/stad3620	
	3. B. Zupancic, C. Gigliotti, A. Duggan, A. H. Caroline Clifford, D. Lown, and J. Baker. A hydrodynamic quantum analogue — walking droplets. <i>Journal of Undergraduate Engineering Physics and Physics Experiments at Queen's</i> , Vol 3. (2022). URL https://ojs.library.queensu.ca/index.php/JUEPPEQ/article/view/15686	
PRESENTATIONS AND POSTERS	• “Poisson Desingularization via Weighted Blowups”.	August, 2026
	• “Poisson Resolution via Weighted Blowups”.	August, 2025
	• “Fuzzy Dark Matter and the Quasi-Particle Hypothesis”	April, 2023
	• “Fuzzy Dark Matter: Dead or Alive?!”	August and October, 2022
CONFERENCES AND WORKSHOPS ATTENDED	• <i>Poisson 2026 Summer School and Conference</i> University of Antwerp and KU Leuven, Belgium.	August, 2026
	• <i>Around Singularities in Poisson Geometry</i> Institute for Advanced Study in Mathematics in Hangzhou, China.	August, 2025
	• <i>Canadian Undergraduate Physics Conference</i> Guelph University in Guelph, ON, Canada.	October, 2022
	• <i>Canadian Astroparticle-Physics Summer Student Talks Competition</i> Laurentian University & SNOLAB in Sudbury, ON, Canada.	August, 2022
SERVICE	Seminar Co-organizer CIRGET <i>Co-organizer of the CIRGET Junior seminar (Centre interuniversitaire de recherches en géométrie et topologie).</i>	Fall 2025 - Present
	Directed Reading Program Mentor McGill University <i>Mentoring undergraduate mathematics reading project.</i>	Winter 2026
	Research and Outreach Fellow Arthur B. McDonald Institute <i>Research project and outreach at a science summer camp.</i>	Summer 2022